

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Developed and evaluated **statistical models** for insurance pricing and fraud risk using **Python** and **scikit-learn**, achieving an **AUC of 0.88** on a held-out dataset of 150,000 insurance applications.
- Engineered **data pipelines** using **Python** and **SQL** to process and merge large datasets from disparate sources, supporting analytical requirements for underwriting and risk assessment.
- Designed and implemented **REST APIs** using **Java Spring Boot** to facilitate data exchange between internal systems and external analytics platforms, handling approximately **2,000 requests per minute**.
- Provided **technical support** and consultation to internal **Product and Sales teams**, translating complex analytical findings into clear, actionable insights for customer presentations.
- Contributed to the research and evaluation of **new analytical techniques** and data mining methodologies, presenting findings and potential applications to the global data science group.
- Collaborated with cross-functional teams to define and implement **data science strategy**, focusing on enhancing current processes and adopting best practices in model development and deployment.
- Containerized **Python-based analytical services** using **Docker** and deployed them onto **AWS EC2** instances, ensuring consistent environments for model training and inference.

SKILLS

Programming Languages – Python, Java, SQL, R

Data Science & ML – Statistical Modeling, Predictive Modeling, Data Mining, Regression Analysis, Classification, Machine Learning, Model Evaluation, Scikit-learn, Pandas, NumPy

Data Processing & ETL – Data Pipelines, Data Warehousing, Data Merging, Data Manipulation, Power Query

Backend & API Development – Java Spring Boot, REST APIs, Microservices, Hibernate, JPA

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, CI/CD, Jenkins, Git

Databases – PostgreSQL, MySQL, Oracle

BI & Visualization – Power BI, Tableau, DAX, Data Visualization

PROJECTS

Insurance Risk Prediction Model Development

- Developed a predictive model using **Python** and **scikit-learn** to assess insurance fraud risk, trained on a dataset of **~220,000 historical claims**.
- Implemented feature engineering techniques to extract relevant variables from raw data, improving model accuracy by **15%** compared to baseline.
- Evaluated model performance using key metrics including **AUC (0.91)** and F1-score, and presented findings to stakeholders for potential product integration.

Enterprise Data Integration and Analytics Platform

- Built **data pipelines** with **Python** and **SQL** to consolidate data from multiple enterprise systems, supporting a unified view for analytics.
- Developed interactive **Power BI dashboards** to visualize key performance indicators for underwriting efficiency and operational metrics, delivering insights to business analysts.
- Managed data transformations and ETL processes using **Power Query** and performed ad-hoc analysis to support business intelligence initiatives.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)